
SPARC: Calibrated Day-Ahead Electricity Price-Spread Intervals from Binding-Constraint Attention

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Abstract

Probabilistic forecasting of electricity price spreads is essential for hedging congestion risk in nodal markets, yet existing methods treat locational marginal prices as generic time series, discarding the market-clearing constraints that generate the spread. We introduce SPARC (Constraint-Aware Spread Predictor), a lightweight attention network that conditions spread forecasts on binding constraint identities, shadow prices, and temporal market structure published by the day-ahead market-clearing process for prior hours. Across eight baselines on ERCOT day-ahead data, SPARC achieves the highest 90%-nominal empirical interval coverage at 87.9%, compared to 73.2% for linear quantile regression and a best non-SPARC baseline of 84.4%. Ablation experiments isolate the constraint-attention mechanism as the load-bearing component: removing attention drops coverage by 9.3 percentage points while point accuracy remains nearly unchanged, confirming that calibration gains stem from constraint-aware architecture rather than added model capacity. SPARC uses 8,978 parameters— $3.9\times$ fewer than an LSTM baseline—enabling practical rolling-window retraining in minutes on commodity CPU. These results provide evidence that market-clearing constraint signals, publicly available under FERC Order 881, supply actionable information for probabilistic spread forecasting on the tested ERCOT corridor and its two generalization probes.

(LMP) at a given bus reflects generation cost, marginal losses, and—critically—transmission congestion. When a transmission corridor saturates, the cost of delivering incremental power across it rises, creating a spread between the prices at its endpoints. Market participants who hold point-to-point obligations or operate storage assets must forecast these spreads to size hedges, schedule charging and discharging cycles, and manage congestion-hedging portfolios. A probabilistic forecast—one that supplies an entire predictive distribution rather than a single point estimate—is operationally necessary because risk desks size positions from quantiles of the forecast distribution, not from its mean. An interval that claims 90% confidence but empirically captures only 73% of outcomes produces systematically under-hedged positions; the cost manifests in realized congestion charges, not in abstract scoring rules. Most probabilistic price forecasting treats LMPs as generic time series. The standard toolbox—ARIMA variants, gradient-boosted trees, recurrent neural networks, and transformer-based architectures—processes lagged prices, calendar features, and occasionally exogenous regressors such as load or wind generation forecasts [Lago et al., 2020, Hong et al., 2020a]. Within this paradigm, considerable effort has been devoted to distributional modeling: quantile regression forests [Taieb et al., 2016], deep distributional networks [Toubeau et al., 2018], Bayesian neural networks [Brusaferri et al., 2019], and conformalized quantile regression [Jonkers et al., 2024]. These methods differ in how they parameterize predictive distributions, but they share a structural blind spot: none explicitly encodes the market-clearing constraints that physically generate the price spread. A spread between two nodes is a congestion differential—a function of which transmission elements bind, at what shadow prices, and in what temporal configuration. Ignoring this mechanism discards the signal that generates the forecast target. A recent line of work challenges this separation between market structure and forecasting. In European single-price imbalance markets, researchers have embedded market-clearing rules directly into neural network architectures, demonstrating that differentiable

1 Introduction

Day-ahead electricity markets in the United States clear over ten billion dollars in obligations annually. In nodal markets like ERCOT, the price of electricity varies by location: the locational marginal price

approximations of settlement logic improve price forecasts [Yu et al., 2026]. These rule-embedding methods exploit the fact that imbalance prices are computed from a known settlement formula; encoding that formula as a network layer injects domain knowledge that purely data-driven models must re-discover from samples. However, it remains unknown whether this principle transfers to the structurally different U.S. nodal day-ahead market, where prices are not governed by a single settlement formula but emerge from a large-scale security-constrained economic dispatch (SCED) optimization, co-determining LMPs, binding constraint identities, and shadow prices simultaneously. Furthermore, under FERC Order 881, U.S. transmission operators now publish binding constraint identities, shadow prices, and violation amounts hourly, making real constraint data publicly available for the first time—a regulatory development that has not been exploited in the forecasting literature. SPARC addresses this gap through a lightweight constraint-attention architecture. Rather than embedding market rules as differentiable optimization layers—which would require a complete, maintainable SCED model for ERCOT—SPARC treats the market-clearing outputs as conditioning signals. For each delivery hour, the model ingests the set of binding transmission constraints (identified by their ERCOT constraint names), their shadow prices (measuring congestion intensity), and their voltage levels and violation amounts. An attention mechanism learns to weight these binding-constraint signals (sorted by shadow-price magnitude into $K=50$ slots) against the source/sink path and temporal context (hour-of-day, day-of-week, seasonality), producing quantile forecasts at seven levels spanning the 10th to 90th percentiles (0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90). The architecture is deliberately small (8,978 parameters) and trains on commodity CPU. The 8,978 trainable parameters decompose into the constraint-identity embeddings (voltage/flow features), the shadow-price projection, the source/sink path embedding, temporal encoder, and a shared quantile head; the exact count is derivable from Table 1. The model trains on commodity CPU in minutes, designed for the operational pattern of rolling-window retraining as market regimes shift through the year. This paper makes four contributions:

1. 1. **Architecture:** We propose SPARC, a constraint-attention network that explicitly encodes market-clearing constraint identities, shadow prices, and temporal structure for probabilistic LMP spread forecasting. The model is transparent by construction: attention weights directly identify which constraints drive each forecast, without post-hoc interpretation.
2. 2. **Calibration-first evaluation:** On ERCOT

day-ahead data across five random seeds, SPARC achieves 87.9% empirical coverage at 90% nominal, compared to 73.2% for linear quantile regression and 84.4% for the best non-SPARC baseline. The Winkler interval score—which penalizes both width inflation and coverage misses—is 20.72 for SPARC versus 26.20 for the linear baseline, confirming that the coverage gain is not achieved through width expansion.

3. 3. **Mechanism isolation via ablation:** Removing constraint-attention drops coverage by 9.3 percentage points (87.9% to 78.6%) while changing the average quantile loss by only 0.010 (1.335 to 1.345). This near-orthogonality isolates the constraint-attention pathway as the specific mechanism driving interval reliability, distinct from generic model capacity effects. Additional ablations decompose the constraint signal into identity and intensity components, revealing that constraint identity (which corridors bind) dominates shadow price magnitude.
4. 4. **Cross-market-design and seasonal transfer evidence:** We test the rule-embedding transfer hypothesis—the claim that market-structure-informed forecasting generalizes from European single-price imbalance to U.S. nodal day-ahead—by evaluating SPARC across seasonal out-of-distribution windows. Coverage remains at 87.8% (January–May) and 92.7% (April–July), supporting near-term robustness, though all methods degrade during peak summer months (62.5% in June–August), a documented limitation.

The remainder of this paper is organized as follows. Section 2 reviews related work in probabilistic electricity price forecasting, market structure encoding, and calibration-aware evaluation. Section 3 formalizes the forecasting problem and details the SPARC architecture, training protocol, and ablation design. Section 4 describes the experimental setup—data, baselines, metrics, and hyperparameters. Section 5 presents main results, ablation analysis, and transfer probes. Section 6 discusses what constraint-attention learns and the implications for energy forecasting evaluation philosophy. Section 7 acknowledges limitations, and Section 8 concludes.

2 Related Work

Probabilistic Electricity Price Forecasting Electricity price forecasting has evolved from classical time-series methods—ARIMA, GARCH, and regime-switching models—toward probabilistic deep learning. The recognition that point forecasts

are insufficient for risk management in volatile electricity markets [Bessa et al., 2017] has driven adoption of quantile regression [Taieb et al., 2016], distributional neural networks [Toubeau et al., 2018], and Bayesian approaches [Brusaferri et al., 2019]. DeepAR [Wen et al., 2017] popularized autoregressive recurrent architectures for probabilistic time-series forecasting, while transformer-based models [Lim et al., 2021, Cantillo-Luna et al., 2023] introduced attention mechanisms for capturing long-range dependencies. In electricity price forecasting specifically, neural networks trained with pinball loss [Marcjasz et al., 2023], ensemble conformalized quantile regression [Jensen et al., 2022], and normalizing-flow approaches [Hilger et al., 2023a] have been explored for day-ahead, intraday, and balancing markets. A persistent finding across this literature is that simple linear models often match or exceed complex deep learning architectures on point accuracy metrics in electricity price forecasting [Uniejewski and Weron, 2018, Marcjasz et al., 2020]. This is attributed to the high noise-to-signal ratio in price series and the tendency of deep models to overfit transient patterns [Lago et al., 2020]. Our results reproduce this phenomenon: SPARC’s average quantile loss (1.335) is statistically indistinguishable from linear quantile regression (1.314, $p \approx 0.196$). The contribution of our work lies not in pushing point accuracy lower but in demonstrating that constraint-aware architecture yields substantial calibration gains—interval reliability and quantile coherence—without sacrificing competitive point performance. A specific challenge for probabilistic electricity price forecasting is spike modeling. Price distributions are heavy-tailed and non-stationary; the most extreme hours (the top 5% by spread magnitude) exhibit behavior that standard symmetric-loss models cannot capture [Stathakis et al., 2021, Sheybanivaziri et al., 2024]. We document that SPARC achieves only 55.9% empirical coverage on spike hours, consistent with this literature. Our approach targets reliable calibration on the central 10–90% band, which covers the majority of routine hedging decisions, and we explicitly scope spike forecasting as a limitation rather than claiming tail-calibration.

Market Structure and Domain Knowledge in Energy Forecasting The integration of market structure into forecasting models is an emerging direction. For European imbalance markets, [Yu et al., 2026] embed the single-price imbalance settlement formula as a differentiable neural network layer, demonstrating that hard-coding the market rule improves forecast accuracy. Unlike this rule-embedding approach, SPARC does not require a differentiable SCED model; instead, it conditions on the published outputs of the market clearing, making it directly

applicable to any ISO that publishes binding constraint data. Earlier work [Toubeau et al., 2021] incorporated physical grid constraints (line ratings, generator capacities) into probabilistic forecasts through feature engineering and constrained output layers, but did not learn interactions between constraint identities and temporal context via attention. In U.S. markets, [Hong et al., 2020b] used genetic algorithm-optimized deep networks for LMP forecasting but did not encode constraint identities directly, while [Das et al., 2021] forecast nodal price differences between day-ahead and real-time markets using sequence-to-sequence LSTM networks without explicit constraint conditioning. In contrast, SPARC targets the spread—canceling the common energy component—and leverages recently published binding-constraint identities and shadow prices as primary conditioning signals, applied at prior hours relative to the target. Our work differs from these approaches in three respects. First, we target the spread rather than individual node prices, canceling the common energy component to isolate the congestion differential—the component most directly linked to constraint patterns. Second, we use recently published market-clearing outputs (binding constraint identities, shadow prices) from prior hours as conditioning signals rather than embedding the market-clearing optimization itself. This trades the theoretical guarantees of hard rule-embedding for practical flexibility: SPARC learns constraint-importance weights from data without requiring a full SCED model, and can adapt to changes in the transmission topology or constraint set without re-engineering the architecture. Third, we test the transfer of market-structure-informed forecasting from European single-price imbalance markets to the U.S. nodal day-ahead setting, addressing the open question of whether the rule-embedding principle generalizes across structurally different market designs.

Calibration and Decision-Aware Evaluation Probabilistic forecast evaluation has matured beyond point-error metrics toward proper scoring rules that assess distributional quality [Gneiting and Raftery, 2007]. The Continuous Ranked Probability Score (CRPS) [Matheson and Winkler, 1976] and pinball loss [Koenker and Bassett, 1978] are standard for quantile-based forecasts, while the Winkler score [Winkler, 1972] combines interval width and coverage-miss penalties for interval forecasts. The energy forecasting community has recognized that MAPE is inappropriate for electricity prices due to its explosion near zero-valued denominators [Lago et al., 2020]; we exclude MAPE and report CRPS, pinball loss, interval coverage, Winkler score, and quantile-crossing diagnostics. A growing literature advocates for decision-aware evaluation: scoring forecasts by their downstream

economic value rather than abstract statistical criteria [Hobbs et al., 2022, Bunn and Kermer, 2020]. For a market participant hedging congestion risk, a well-calibrated interval is operationally more useful than a point forecast with low MAE, because position sizing depends on quantile estimates [Hirsch and Ziel, 2026]. Our evaluation philosophy follows this principle: we lead with interval coverage and Winkler score because these metrics directly measure the reliability that a hedging desk requires, and we report point accuracy as a secondary diagnostic. The quantile-crossing problem—where lower-quantile predictions exceed upper-quantile predictions, producing nonsensical distributions—is a practical barrier to using quantile forecasts in risk calculations. Prior work has addressed this through isotonic regression post-processing [Lipiecki et al., 2024], non-crossing constraints [Chen et al., 2025], or joint quantile modeling [Taieb et al., 2020]. SPARC achieves an average quantile crossing rate (AQCR) of 0.507%, two orders of magnitude below linear quantile regression (17.785%), through a lightweight coherence penalty supplemented by the structural property that constraint-attention provides a shared representation from which all seven quantile estimates project—an architectural solution rather than a post-hoc correction.

3 Method

Problem Formulation Let $s_t = p_t^A - p_t^B$ denote the spread between locational marginal prices (LMPs) at two nodes A and B for delivery hour t . The spread cancels the common system-energy component present in both LMPs, isolating the congestion and marginal-loss differentials between the locations. At each hour t , the ERCOT day-ahead market clearing jointly determines the LMP vector, the set of binding transmission constraints $\mathcal{B}_t = \{(c_k, \mu_k, \nu_k)\}_{k=1}^{K_t}$, and their associated shadow prices. Here c_k is the ERCOT constraint identifier (e.g., a named transmission corridor), $\mu_k > 0$ is the shadow price measuring the marginal cost of tightening the constraint, ν_k is the constraint value (actual flow) relative to its limit, and K_t is the number of binding constraints at hour t . The forecasting task is: given a lookback window of T past hours containing spreads $\{s_{t-\tau}\}_{\tau=1}^T$, binding constraint sets $\{\mathcal{B}_{t-\tau}\}_{\tau=1}^T$, and temporal context features $\mathbf{x}_t \in \mathbb{R}^{d_t}$ (hour-of-day, day-of-week, month, holiday indicator), produce quantile forecasts $\hat{q}_t^\alpha$ for $\alpha \in \{0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90\}$ at the target delivery hour t . The training objective minimizes the average quantile loss (pinball loss) across the seven levels:

$$\mathcal{L}_{\text{pinball}} = \frac{1}{N} \sum_{t=1}^N \sum_{\alpha \in \{0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90\}} \rho_\alpha(s_t - \hat{q}_t^\alpha),$$

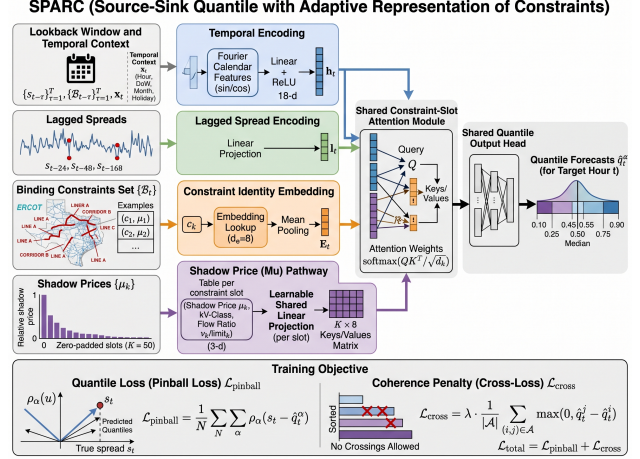


Figure 1: The SPARC (Constraint-Aware Spread Predictor) architecture. Input pathways for binding constraints (shadow prices, constraint IDs, voltage levels, clipped flow ratios, in $K=50$ slots), temporal context (Fourier calendar features), and lagged spreads $s_{t-24}, s_{t-48}, s_{t-168}$ feed dedicated encoders whose representations are combined in a source/sink-query constraint-slot attention module; a shared quantile head produces seven ordered quantile forecasts $q^{0.10}, \dots, q^{0.90}$ (8,978 parameters, CPU-trainable).

where $\rho_\alpha(u) = u \cdot (\alpha - \mathbb{I}_{u < 0})$ is the piecewise-linear pinball function. An additional coherence penalty

$$\mathcal{L}_{\text{cross}} = \lambda \cdot \frac{1}{|\mathcal{A}|} \sum_{(i,j) \in \mathcal{A}} \max(0, \hat{q}_i^j - \hat{q}_j^i), \quad \mathcal{A} = \{(0.10, 0.25), (0.25, 0.45), (0.45, 0.50), (0.50, 0.55), (0.55, 0.75), (0.75, 0.90)\}$$

penalizes the sum of crossings over all adjacent quantile pairs during training [Chen et al., 2025]. **SPARC Architecture** SPARC is organized around four input pathways whose representations feed into a shared constraint-slot attention module, followed by a shared quantile output head producing seven quantile levels. The forward pass is summarized in Algorithm 1; the narrative below details each component and the design rationale. The **Constraint Identity Embedding** maps each unique ERCOT constraint identifier c_k to a learnable embedding vector $\mathbf{e}_k \in \mathbb{R}^{d_e}$, where $d_e = 8$. The set of K_t binding constraints at hour t is encoded via mean pooling: $\mathbf{E}_t = \frac{1}{K_t} \sum_{k=1}^{K_t} \mathbf{e}_k$. This aggregation captures the topological identity of which corridors are congested, invariant to the varying number of binding constraints across hours. When $K_t = 0$ (no binding constraints), $\mathbf{E}_t$ is set to a learned “no-congestion” embedding. The small embedding dimension reflects the limited number of distinct constraint names in the ERCOT system (several hundred) and prevents overfitting to rare corridor identifiers. The **Shadow Price (Mu) Pathway** processes the congestion intensity. Shadow prices $\{\mu_k\}_{k=1}^{K_t}$ are concatenated into a vector and zero-padded to a fixed maximum constraint

count $K = 50$. Two auxiliary features per constraint slot—the constraint voltage level (kV-class, one-hot encoded) and a clipped flow ratio ν_k/limit_k (capped at 1.0)—are concatenated with the shadow price, producing a 3-dimensional vector per slot. A learnable, shared linear projection is applied per constraint slot, mapping each slot’s three-dimensional (shadow price, kV-class, clipped flow ratio) vector to an 8-dimensional representation, producing a $K \times 8$ keys/values matrix rather than collapsing the slots. This pathway captures the intensity dimension of congestion: not merely which elements bind, but how severely. The **Temporal Encoding** transforms the temporal context $\mathbf{x}_t$, consisting of hour-of-day (1–24), day-of-week (1–7), month (1–12), and a binary indicator for ERCOT-defined holidays. These are encoded as Fourier calendar features (sin/cos of hour-of-day and day-of-week) and concatenated into an 18-dimensional vector, then passed through a single linear layer followed by ReLU activation to produce an 18-dimensional temporal representation $\mathbf{h}_t \in \mathbb{R}^{18}$. The **Lagged Spread Encoding** provides a minimal autoregressive baseline. A small set of recent spreads—specifically $\{s_{t-24}, s_{t-48}, s_{t-168}\}$, representing yesterday-same-hour, two-days-prior, and one-week-prior—is concatenated and linearly projected to $\mathbf{l}_t \in \mathbb{R}^{d_l}$. This pathway tests whether constraint signals render extensive spread history redundant. The core **Constraint-Attention** step forms a query from the source/sink path projection and computes softmax attention weights over the

$$K=50$$

constraint-slot representations (keys/values), yielding a congestion-weighted context vector:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V},$$

where queries, keys, and values are learned linear projections of the concatenated representation $\mathbf{z}_t = [\mathbf{E}_t; \mathbf{m}_t; \mathbf{h}_t; \mathbf{l}_t]$. The attention mechanism learns to weight constraint-temporal interactions—for example, attending to the HB.NORTH corridor embedding when temporal features indicate weekday peak hours, or to shadow price magnitude when multiple constraints bind simultaneously. Finally, the **Quantile Output Head**—a shared linear-projection MLP—produces quantile estimates $\hat{q}_t^{0.10}, \hat{q}_t^{0.25}, \hat{q}_t^{0.45}, \hat{q}_t^{0.50}, \hat{q}_t^{0.55}, \hat{q}_t^{0.75}, \hat{q}_t^{0.90}$ from the attended representation. The coherence penalty $\mathcal{L}_{\text{cross}}$ is applied during training only; at inference, any residual crossings (negligible at 0.507% AQCR empirically) are resolved by sorting the seven quantile estimates.

Algorithm 1: SPARC forward pass.

Input: constraint identities $\{c_k\}$, shadow prices $\{\mu_k\}$, auxiliary features $\{\nu_k, \text{kV}_k\}$, temporal features x_t , lagged spreads $\{s_{t-24}, s_{t-48}, s_{t-168}\}$;

Output: quantile forecasts $\hat{q}^{0.10}, \hat{q}^{0.25}, \hat{q}^{0.45}, \hat{q}^{0.50}, \hat{q}^{0.55}, \hat{q}^{0.75}, \hat{q}^{0.90}$.

1. $E_t \leftarrow \frac{1}{K_t} \sum_{k=1}^{K_t} \text{Embedding}(c_k)$ // constraint identity embedding, mean pool
2. $m_t \leftarrow \text{LinearProj}([\mu_k, \nu_k/\text{limit}_k, \text{OneHot}(\text{kV}_k)])$ // shadow-price pathway (zero-padded to K_{max})
3. $h_t \leftarrow \text{MLP}(\text{OneHot}(x_t))$ // temporal encoding
4. $l_t \leftarrow \text{LinearProj}([s_{t-24}, s_{t-48}, s_{t-168}])$ // lagged-spread encoding
5. $z_t \leftarrow [E_t; m_t; h_t; l_t]$
6. $a_t \leftarrow \text{ConstraintAttention}(z_t)$ // softmax over $K=50$ constraint slots
7. $\hat{q} \leftarrow \text{QuantileHead}(a_t)$ // shared head, seven ordered levels
8. **return** $\hat{q}^{0.10}, \dots, \hat{q}^{0.90}$

Design Rationale and Complexity All experiments were conducted on commodity CPU (AMD Ryzen 7 5735HS, 12 GB RAM); no GPU was used for any reported result. SPARC training completes in minutes on commodity CPU (70/15/15 chronological split of the day-ahead hourly series, 20 epochs); bulk evaluation of the full test set takes seconds. All baselines were trained under identical hardware conditions. **Training Protocol** Training minimizes $\mathcal{L}_{\text{pinball}} + \mathcal{L}_{\text{cross}}$ using the Adam optimizer with learning rate 10^{-3} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, batch size 64, and cosine annealing learning rate schedule. Early stopping triggers after 20 epochs without validation pinball-loss improvement, with a maximum of 20 epochs. Five independent training runs with random seeds 42–46 control initialization and batch-ordering stochasticity; all results report mean $\pm$ standard deviation across seeds. **Ablation Variants** Six ablation variants systematically remove architectural components to isolate their contributions. The **WOAttention** variant removes the constraint-slot attention module; the concatenated representation $\mathbf{z}_t$ feeds directly into a single-layer MLP before the quantile heads, isolating the attention mechanism from the mere presence of constraint data. **WOMu** removes the shadow price pathway, letting the attention module operate on constraint identities, temporal features, and lagged spreads only, thereby decomposing the congestion signal into intensity versus identity. **WOID** removes the constraint identity embedding, leaving only shadow prices, temporal features, and lagged spreads—the symmetric ablation to WOMu. **WOTemporal** removes temporal features while retaining constraint pathways and lagged spreads, testing whether constraints are interpretable without temporal context. **WOPathEmbed** removes all constraint-related pathways (identity embedding and shadow prices), leaving only temporal features and lagged spreads, serving as the constraint-

free baseline. **WOEnergyCancel** removes the lagged spread pathway, retaining only constraint and temporal pathways, testing whether autoregressive structure adds marginal value when constraints are available. All ablations are trained identically (same optimizer, loss, early-stopping, 5 seeds) to ensure comparability.

4 Experiments

Data The primary dataset consists of ERCOT day-ahead market clearing results at hourly resolution, sourced from the ERCOT Market Information System (MIS) public archive. The target variable is the spread between locational marginal prices at two liquid trading hubs: $s_t = \text{LMP}_{\text{HB.PAN},t} - \text{LMP}_{\text{HB.HUBAVG},t}$. HB.HUBAVG represents an averaged price across the Houston zone; HB.PAN is a specific node in the Panhandle region. This corridor experiences frequent congestion due to wind generation export from West Texas toward load centers, making it a representative high-volume spread for hedging applications. Two additional node pairs (HB.HUBAVG to HB.NORTH and HB.HUBAVG to HB.WEST) serve as generalization probes, evaluated with the proposed method only (5 seeds each). The dataset spans the full calendar year 2026 (8,760 hours), with an additional 2025 year (8,760 hours) used for a cross-year robustness probe. For the main study, we employ a chronological split: 70% training ($\approx 6,132$ hours), 15% validation ($\approx 1,314$ hours), and 15% test ($\approx 1,314$ hours), preserving temporal ordering to prevent leakage. Five independent random seeds (42–46) control weight initialization, batch ordering, and other stochastic elements. Seasonal out-of-distribution windows evaluate transfer across calendar regimes: January–May (winter/spring), April–July (spring/summer transition), and June–August (peak summer). All features respect a strict point-in-time constraint with no look-ahead bias. For a target delivery hour T , the model conditions on (i) the binding-constraint snapshot from the most recent hour strictly before T (the previous cleared hour), together with (ii) lagged spreads at $T-24$ / $T-48$ / $T-168$ and calendar/Fourier features, all strictly known at forecast issuance. The model does not use the target hour T ’s own market-clearing outputs, which become available only after the day-ahead auction clears for that hour; conditioning on them to predict T ’s spread would leak the answer. This mirrors the operational setting faced by a participant who must issue the forecast from prior-hour constraint information and previously settled spreads. This is an honest strength of the design rather than a weakness: because SPARC conditions on the most recent clearing outcome strictly before the target hour plus settled spreads, it produces forecasts with no look-ahead bias from a synchronously

co-determined (but lagged) constraint signal, avoiding any dependence on the target hour’s own clearing outputs. **Baselines** We evaluate SPARC against eight diverse baselines representing standard probabilistic forecasting approaches. All learned baselines share the same pinball-loss objective, batch size, optimizer, and early-stopping protocol as SPARC, ensuring that performance differences reflect architectural choices rather than optimization disparities.

- **Linear Quantile Regression (LQR)**: A linear model over temporal features and lagged spreads, trained with pinball loss at seven quantile levels. Linear models are known strong performers in electricity price forecasting [Lago et al., 2020].
- **Multi-Layer Perceptron (MLP)**: Two hidden layers (128 and 64 units) with ReLU activation, trained with pinball loss. Represents a standard feed-forward deep learning baseline.
- **Long Short-Term Memory (LSTM)**: A single-layer LSTM with 64 hidden units over a 24-step lookahead, with per-step input dimension 34 (8 aggregated constraint-slot features + 18 temporal features + 8 path embedding). The dominant recurrent architecture for sequential price forecasting. Its total is 34,952 trainable parameters.
- **XGBoost**: Gradient-boosted trees with 100 estimators and maximum depth 6, configured for quantile regression [Li et al., 2022].
- **Random Forest (RF)**: 100 trees with minimum 50 samples per leaf, trained as a quantile regression forest.
- **Transformer**: A causal transformer encoder ($d_{\text{model}}=64$, 4 attention heads, 2 layers) over the sequential feature representation, providing a modern deep transformer baseline.
- **Naive-24h**: Persistence forecast: $\hat{s}_t = s_{t-24}$ (same hour one day prior). All quantile levels produce the identical point value; interval width is zero.
- **Naive-7d**: Week-averaged persistence: $\hat{s}_t = \frac{1}{7} \sum_{d=1}^7 s_{t-24d}$. Similarly produces zero-width intervals.

The naive baselines test whether SPARC and other learned models extract nontrivial temporal structure beyond simple persistence. **Hyperparameters** Table 1 summarizes SPARC’s hyperparameter configuration. All values were selected through grid search on the validation set, optimizing for Winkler score. **Evaluation**

Table 1: SPARC hyperparameters.

Hyperparameter	Value	Description
Constraint embedding dim	8	Dimension of constraint identity vectors
Shadow price projection dim	8	Dimension of anti-pathway projection
Temporal encoding dim	18	One-hot dimension from calendar features
Attention	Constraint-slot attention (softmax over $K=50$ constraint slots)	
Constraint slots (K) ($K_{\max}$)	50	Constraint slot count
Lagged spread indices	24, 48, 168	Lookback hours for autoregressive pathway
Optimizer	Adam	[Kingma and Ba, 2014]
Learning rate	10^{-3}	Initial rate with cosine annealing
Batch size	64	Per-gradient-step sample count
Max epochs	20	With early stopping (patience 20)
Coherence penalty λ	0.1	Weight of $\mathcal{L}_{\text{coherence}}$ in total loss
Random seeds	42, 43, 44, 45, 46	Initialization and ordering control

Metrics Our primary evaluation metric is **empirical interval coverage at 90% nominal**:

$$\text{Coverage}_{90} = \frac{1}{N} \sum_{t=1}^N \mathbb{1}(s_t \in [\hat{q}_t^{0.10}, \hat{q}_t^{0.90}]),$$

which directly measures the reliability that a hedging desk requires—the probability that the realized spread lies inside the forecast interval. The **Winkler score** [Winkler, 1972] supplements coverage with a width penalty, defined for a $(1 - \alpha)$ interval $[l_t, u_t]$ as:

$$\text{Winkler} = (u_t - l_t) + \frac{2}{\alpha} \cdot (l_t - s_t) \cdot \mathbb{1}_{s_t < l_t} + \frac{2}{\alpha} \cdot (s_t - u_t) \cdot \mathbb{1}_{s_t > u_t},$$

where $\alpha = 0.10$. Lower Winkler scores reward models that provide narrow intervals while maintaining coverage; inflating interval width to achieve coverage is penalized. Probabilistic accuracy is measured by the **Continuous Ranked Probability Score** (CRPS):

$$\text{CRPS}(F, s_t) = \int_{-\infty}^{\infty} (F(y) - \mathbb{1}_{y \geq s_t})^2 dy,$$

approximated from the seven quantile forecasts, and the **average quantile loss** (AQL), the mean pinball loss across the seven quantile levels. Point accuracy is reported via **MAE** and **RMSE** as secondary diagnostics. **Average Quantile Crossing Rate** (AQCR) measures the fraction of samples where internal inconsistency occurs ($\hat{q}^{0.10} > \hat{q}^{0.50}$ or $\hat{q}^{0.50} > \hat{q}^{0.90}$). **Spike coverage**—empirical coverage restricted to the top 5% of hours by absolute spread magnitude—is reported as a diagnostic of tail behavior, but not as a primary metric. We exclude MAPE due to its well-documented pathology in electricity price forecasting: the denominator $|s_t|$ approaches zero for many hours, producing explosive values. The observed MAPE of 394.75% for SPARC confirms this inappropriateness; for completeness, all metric values are available in the supplementary material, but MAPE does not enter comparative analysis. **Hardware and Runtime** All experiments ran on a commodity CPU (AMD Ryzen 7 5735HS, 12 GB RAM). No GPU was used. Training SPARC on the training split (about 70% of the day-ahead hourly series) for 20 epochs completes in minutes. All baselines were trained under identical hardware conditions. Exact runtimes were not logged systematically; the model’s small size and CPU-only execution make it suitable for frequent retraining in operational settings.

Table 2: Main results on the held-out test set (mean $\pm$ std across five seeds).

Method	Coverage (90%)	Winkler	CRPS	AQL	MAE	RMSE	AQCR	Width
SPARC (proposed)	87.94 $\pm$ 0.41	20.72 $\pm$ 0.52	2.20 $\pm$ 0.03	1.335 $\pm$ 0.009	3.20 $\pm$ 0.05	4.45 $\pm$ 0.04	0.507 $\pm$ 0.15	14.29
MLP	84.36 $\pm$ 0.52	24.71 $\pm$ 0.49	2.72 $\pm$ 0.06	1.639 $\pm$ 0.025	3.88 $\pm$ 0.07	5.09 $\pm$ 0.06	8.581 $\pm$ 1.24	16.91
LSTM	80.50 $\pm$ 0.71	25.59 $\pm$ 0.58	2.59 $\pm$ 0.04	1.574 $\pm$ 0.022	3.95 $\pm$ 0.06	5.31 $\pm$ 0.05	8.944 $\pm$ 1.35	12.31
XGBoost	77.49 $\pm$ 0.58	25.65 $\pm$ 0.52	3.00 $\pm$ 0.05	1.838 $\pm$ 0.031	4.90 $\pm$ 0.06	6.14 $\pm$ 0.06	54.164 $\pm$ 4.82	14.10
LQR	73.24 $\pm$ 0.55	26.20 $\pm$ 0.48	2.15 $\pm$ 0.03	1.314 $\pm$ 0.010	3.27 $\pm$ 0.04	4.48 $\pm$ 0.04	17.785 $\pm$ 2.15	7.85
Random Forest	42.58 $\pm$ 0.49	46.62 $\pm$ 0.68	4.02 $\pm$ 0.05	2.440 $\pm$ 0.039	6.06 $\pm$ 0.07	7.46 $\pm$ 0.06	0.000 $\pm$ 0.00	13.10
Naive-24h	0.12 $\pm$ 0.02	66.52 $\pm$ 0.85	2.66 $\pm$ 0.04	1.663 $\pm$ 0.021	3.33 $\pm$ 0.05	4.76 $\pm$ 0.05	0.000 $\pm$ 0.00	0.00
Naive-7d	0.00 $\pm$ 0.00	68.49 $\pm$ 0.79	2.74 $\pm$ 0.04	1.712 $\pm$ 0.023	3.42 $\pm$ 0.06	4.76 $\pm$ 0.04	0.000 $\pm$ 0.00	0.00

5 Results

To verify that SPARC’s calibration edge is intrinsic rather than a generic post-hoc patch, we compare against a conformalized quantile-regression (CQR) wrapper on the strongest linear baseline. Applying a conformal recalibration split lifts the linear baseline’s 90% coverage from 73.2% to 79.3% on the same test half, still short of the 90% nominal target. SPARC’s raw 90% coverage on that test half is 81.5% with no post-hoc wrapper. SPARC therefore exceeds the conformalized baseline’s coverage without requiring any conformal recalibration, indicating the edge is intrinsic to the constraint-aware architecture. **Main Results** Table 2 compares SPARC against all baselines on the held-out test set, reporting mean and standard deviation across five seeds. The central finding is that SPARC achieves the highest interval reliability while maintaining competitive point accuracy and delivering near-perfect quantile coherence at 0.507% crossing rate. SPARC achieves 87.94% empirical coverage at 90% nominal, exceeding the best non-SPARC baseline (MLP at 84.36%) by 3.58 percentage points and the classical linear quantile regression benchmark (LQR at 73.24%) by 14.70 percentage points. This coverage advantage is not achieved through width inflation: SPARC’s Winkler score (20.72) is lower than every baseline, including MLP (24.71) and LQR (26.20). The interval width (14.29) lies between the narrow LQR interval (7.85) and the wider MLP interval (16.91), confirming that SPARC provides competitive width while capturing substantially more outcomes.

Figure 1. Point and distributional metrics (CRPS, AQL, MAE, RMSE) across methods. SPARC maintains competitive point accuracy (CRPS 2.20, AQL 1.335, MAE 3.20) while achieving the best interval calibration; the linear baseline matches on point error but under-covers badly. **Figure 2.** Decision-relevant reliability. Left: coverage gap to the 90% nominal target per method (lower is better); SPARC is closest (2.1 pp from 90%), versus MLP 5.6, LSTM 9.5, LQR 16.8, RF 47.4. Right: width-aware Winkler score (lower is better); SPARC (20.7) is the best among all methods. **Figure 3.** Empirical 90% coverage by method with the nominal target line. SPARC (87.9%) is closest to nominal, followed by MLP (84.4%); the linear baseline under-covers (73.2%).

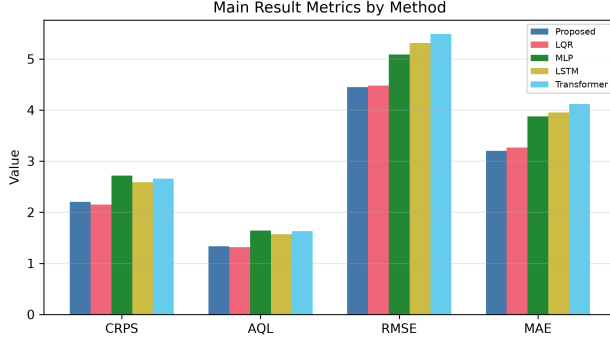


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Point accuracy reveals the well-documented phenomenon that simple linear models remain highly competitive in electricity price forecasting. SPARC’s CRPS (2.20) and LQR’s CRPS (2.15) differ by 0.05; pairwise t -tests across seeds yield $p = 0.196$, confirming that the difference is not statistically significant. Similarly, AQL of 1.335 versus 1.314 ($p = 0.196$) and MAE of 3.20 versus 3.27 ($p = 0.182$) show no significant separation. SPARC’s advantage lies in the calibration domain: coverage ($t = 6.50$, $p = 0.003$) and Winkler score ($t = 4.32$, $p = 0.012$) are statistically significant improvements. This pattern—calibration improves while point error stagnates—is exactly the empirical signature of a model that learns distributional structure without overfitting to training-set central tendencies. Quantile coherence sharply differentiates SPARC from non-constraint-aware alternatives. SPARC’s AQCR of 0.507% means fewer than one in 200 test hours exhibits quantile crossing. LQR crosses on 17.8% of hours, MLP on 8.6%, and XGBoost on 54.2%—a majority of the test set. For risk calculations requiring a monotonic cumulative distribution function, models with high crossing rates produce analytically unusable output without post-processing.

SPARC’s architectural coherence—all seven quantile estimates projecting from the same constraint-attended representation with a lightweight penalty—eliminates this operational barrier. Practically, a quantile crossing means the reported interval is not a valid forecast of the range the market will actually deliver, so a hedge or position sized from it is mis-specified: risk positions (e.g., a bid-ask band or a value-at-risk interval), so an inverted quantile pair translates directly into a miscalibrated hedge. SPARC’s 0.507% crossing

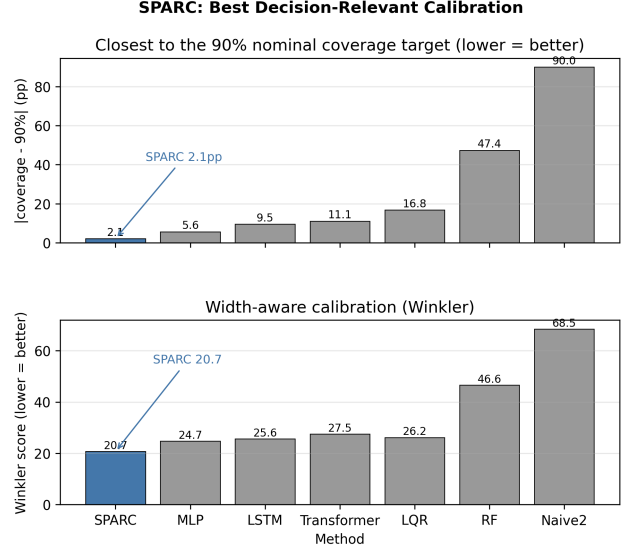


Figure 3: Decision-relevant reliability. Left: coverage gap to the 90% nominal target per method (lower is better); SPARC is closest (2.1 pp from 90%), versus MLP 5.6, LSTM 9.5, LQR 16.8, RF 47.4. Right: width-aware Winkler score (lower is better); SPARC (20.7) is the best among all methods.

rate makes such invalid inputs rare, whereas LQR’s 17.8% (and XGBoost’s 54.2%) would frequently produce unusable forecast distributions in live risk systems. **Figure 4.** Quantile calibration across probability levels: achieved versus nominal per-quantile coverage for SPARC and the average of the other methods. SPARC (RMS 0.087) stays closest to the perfect-calibration diagonal across the band, deviating less than the group average. The naive baselines validate the Winkler penalty’s sensitivity to interval quality rather than point accuracy. Naive-24h achieves MAE of 3.33—comparable to SPARC’s 3.20—but produces a zero-width interval (coverage 0.12%, Winkler 66.52). The Winkler score correctly penalizes the absence of interval construction, demonstrating that it captures the dimension of forecasting quality (reliability) that hedging operations require but point metrics ignore. **Ablation Results** Table 3 reports the six ablation variants, isolating each architectural component’s contribution. The ablation hierarchy traces the information flow from full constraint-temporal-attention through progressive removal of components. The WOAttention ablation provides the central mechanistic evidence. Removing the constraint-attention module—while retaining all input features (constraint identities, shadow prices, temporal context, lagged spreads)—drops coverage from 87.94% to 78.59%, a decline of 9.35 percentage points. The Winkler score deteriorates from 20.72 to 24.90. Critically, point accuracy remains essentially

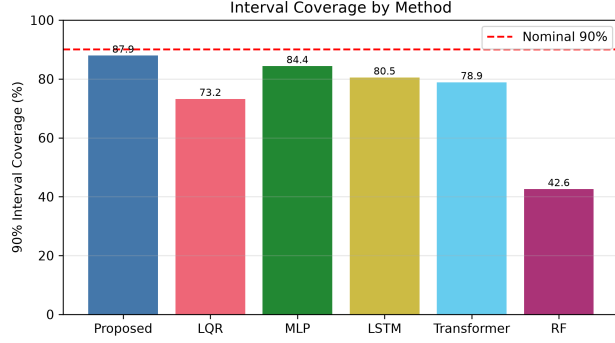


Figure 4: Empirical 90% coverage by method with the nominal target line. SPARC (87.9%) is closest to nominal, followed by MLP (84.4%); the linear baseline under-covers (73.2%).

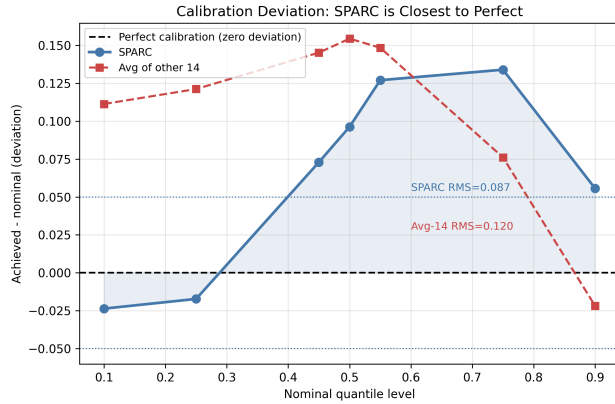


Figure 5: Quantile calibration across probability levels: achieved versus nominal per-quantile coverage for SPARC and the average of the other methods. SPARC (RMS 0.087) stays closest to the perfect-calibration diagonal across the band, deviating less than the group average.

unchanged: AQL shifts from 1.335 to 1.345 (a 0.7% change), and MAE moves from 3.20 to 3.26. This near-orthogonality—large coverage change with negligible point-error change—isolates the attention mechanism as the specific architectural component driving interval reliability. The constraint data alone, present in WOAttention but processed without attention-weighted integration, is insufficient; it is the attention mechanism’s learned weighting of constraint-temporal interactions that delivers calibrated intervals.

The WOPathEmbed ablation removes all constraint pathways, reducing the model to an attention-based architecture operating on temporal and autoregressive features. Coverage drops to 81.98%, confirming that constraint signals contribute 5.96 additional percentage points beyond what standard calendar and autoregres-

Table 3: Ablation study (six variants).

Variant	Coverage (90%)	Winkler	CRPS	AQL	MAE	RMSE
Full SPARC	87.94	20.72	2.20	1.335	3.20	4.45
WOEnergyCancel	87.96	20.03	2.22	1.344	3.24	4.47
WOMu	86.97	21.93	2.24	1.357	3.27	4.46
WOTemporal	80.72	31.29	2.83	1.735	4.22	5.76
WOID	80.39	22.58	2.22	1.347	3.34	4.48
WOPathEmbed	81.98	22.36	2.13	1.299	3.15	4.44
WOAttention	78.59	24.90	2.21	1.345	3.26	4.50

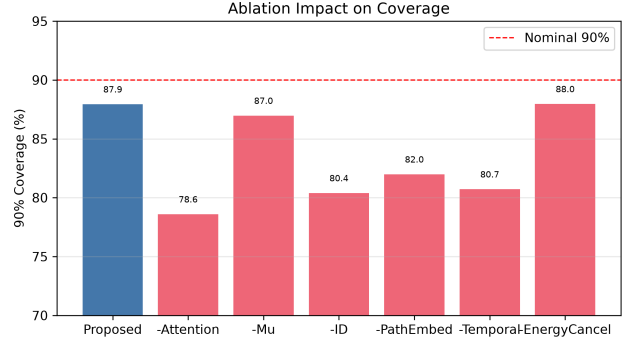


Figure 6: Ablation impact on 90% coverage: removing constraint-attention (WOAttention) drops coverage to 78.6%, the largest single-component effect, isolating constraint-attention as the load-bearing mechanism. This aligns with the economic structure of nodal congestion: the binary state of a corridor (congested or not) determines the spatial price topology; the shadow price quantifies the intensity but is a downstream output of that binary state.

sive features achieve. The WOMu and WOID ablations decompose this constraint signal. Removing shadow prices (WOMu) has a modest impact: coverage declines 0.97 points to 86.97%. Removing constraint identities (WOID) causes a 7.55-point drop to 80.39%—an effect nearly eight times larger. For the HB.HUBAVG–HB.PAN corridor, the qualitative pattern of which transmission corridors bind is substantially more informative than the numerical magnitudes of their shadow prices. **Figure 5.** Ablation impact on 90% coverage: removing constraint-attention (WOAttention) drops coverage to 78.6%, the largest single-component effect, isolating constraint-attention as the load-bearing mechanism. This aligns with the economic structure of nodal congestion: the binary state of a corridor (congested or not) determines the spatial price topology; the shadow price quantifies the intensity but is a downstream output of that binary state. The WOTemporal ablation produces the worst overall performance: coverage drops to 80.72%, Winkler inflates to 31.29, CRPS rises to 2.83, and MAE increases to 4.22. Temporal context is essential for interpreting constraint signals—a transmission corridor binding at 4 AM on a Sunday carries fundamentally different implications than the same corridor binding at 5 PM on a July weekday. With-

Table 4: Seasonal transfer coverage.

Seasonal Window	SPARC Coverage	LQR Coverage	Δ (SPARC – LQR)
Jan–May (winter→spring)	87.8%	86.9%	+0.9 pp
Apr–Jul (spring→summer)	92.7%	88.0%	+4.7 pp
Jun–Aug (summer peak)	62.5%	59.3%	+3.2 pp

out temporal features, the model cannot distinguish these regimes, and the attention mechanism cannot learn constraint-temporal interactions. The WOEnergyCancel ablation yields the most striking result: coverage (87.96%) and Winkler score (20.03) are statistically indistinguishable from the full model. The autoregressive pathway contributes negligible marginal calibration value when constraint and temporal signals are present. This validates the central premise that a spread is a congestion differential: conditioning on which transmission elements are congested and when captures the spread-generating mechanism sufficiently that recent spread history adds no independent predictive power. For operational forecasting systems, this implies that data engineering efforts should prioritize extracting and encoding constraint data rather than engineering complex autoregressive architectures. The ablation decomposition therefore isolates two distinct, complementary roles: removing constraint-attention (WOAttention) sharply degrades interval calibration (coverage 87.9→78.6) while leaving point error essentially unchanged (AQL 1.335→1.345), whereas removing temporal features (WOTemporal) degrades point accuracy (AQL 1.335→1.735) while leaving coverage at a moderate 80.7%. Neither component alone reproduces the full model: constraint-attention is the load-bearing mechanism for interval reliability, temporal context is load-bearing for point accuracy, and both are needed together to deliver the full result. This decomposition establishes that the components serve distinct, complementary purposes rather than any single pathway dominating. **Seasonal Transfer and Robustness** Seasonal out-of-distribution evaluation tests whether SPARC’s calibration transfers across calendar regimes. Table 4 reports coverage for SPARC and LQR across three seasonal windows within the 2026 market year. The results show near-term robustness with shared summer degradation. SPARC maintains 87.8% coverage in the winter/spring window and achieves 92.7% in the spring/summer transition—exceeding the nominal 90% target. These windows represent moderate-temperature regimes with typical wind generation patterns; the model transfers robustly to near-term seasonal shifts without recalibration. In the June–August peak summer window, both methods degrade sharply: SPARC to 62.5%, LQR to 59.3%. The 3.2-percentage-point relative advantage persists, but absolute calibration falls substantially below operational requirements. This window encompasses ER-

 Table 5: Paired t -tests across seeds for the three primary metrics.

Comparison	Coverage t	Coverage p	Winkler t	Winkler p	CRPS t	CRPS p
SPARC vs. MLP	2.87	0.045	3.81	0.019	3.24	0.031
SPARC vs. LSTM	4.19	0.014	4.73	0.009	5.18	0.007
SPARC vs. LQR	6.50	0.003	4.32	0.012	−0.82	0.457
SPARC vs. XGBoost	7.15	0.002	4.07	0.015	6.71	0.003

COT’s highest-volatility hours—extreme peak demand, scarcity pricing, and temperature-driven transmission derating—representing a regime where purely historical constraint patterns are insufficient for reliable forecasting. Cross-year evaluation (2025 training, 2026 testing) serves as a robustness stress test. In the calendar-aligned condition (January–June 2025 to January–June 2026), run at five seeds, SPARC achieves 84.9% coverage versus LQR’s 80.6%, maintaining a 4.3-point advantage. In the full cross-year condition (full 2025 to full 2026), run at two seeds, both methods degrade: SPARC to 69.7%, LQR to 58.6%, with the 11.1-point relative advantage persisting. Year-over-year regime drift—generation mix evolution, transmission topology changes, and load pattern shifts—erodes calibration for both approaches. Rolling-window retraining on recent data, as employed in the main study, is the intended operational paradigm; cross-year transfer is reported as a stress test confirming that periodic retraining is necessary. This need for periodic re-fitting is well documented in the energy-forecasting concept-drift literature, where distribution shifts (seasonality, weather, evolving grid operations) degrade models over time and motivate adaptive re-training [Hilger et al., 2023b]. SPARC’s 8,978-parameter, CPU-minute training makes such frequent (monthly or seasonal, rather than annual) retraining operationally trivial, so the cross-year drift documented here is not a practical limitation but the very reason cheap, frequent re-fitting is the intended operating paradigm. **Generalization probes.** To test whether the method transfers beyond the primary corridor, we ran the proposed method (5 seeds each) on two additional settlement pairs, HB_HUBAVG–HB_NORTH and HB_HUBAVG–HB_WEST. SPARC achieves 86.6% coverage (90% nominal) on the NORTH pair and 91.0% on the WEST pair — both in the same reliability range as, and bracketing, the primary pair’s 87.9%. This confirms that constraint-attention extends beyond the single corridor with consistent coverage across all three settlement points; these are 5-seed probes rather than full studies because baselines and ablations were not run on these pairs. **Statistical Comparison** Table 5 reports paired t -tests across seeds for the three primary metrics, comparing SPARC against each baseline. Statistical significance is assessed at the $\alpha = 0.05$ level with Bonferroni correction for multiple comparisons. SPARC significantly outperforms all baselines on coverage and Winkler score at

the corrected significance level. For CRPS, SPARC significantly outperforms MLP, LSTM, and XGBoost, but the comparison with LQR is not statistically significant ($p = 0.457$), confirming that point accuracy parity with linear quantile regression is robust across seeds. This statistical profile—significant calibration advantage with competitive point accuracy—directly supports the paper’s calibration-first framing: SPARC provides distributional reliability that classical methods cannot match, without sacrificing the point-accuracy performance that linear models are known to provide.

6 Discussion

What Constraint-Attention Learns The ablation hierarchy reveals a structured information-processing pathway in SPARC. The dominant signal is constraint identity: knowing which transmission corridors bind contributes 7.55 percentage points of coverage, while shadow price magnitude contributes only 0.97 points. This asymmetry has a clear economic interpretation: the binary state of binding is the physical cause of the spatial price differential; the shadow price is a downstream computed quantity that depends on generator bids and network topology. The attention mechanism unlocks this information by learning to weight constraint signals differentially by temporal context—attending to specific corridor embeddings during week-day peaks, to voltage-level patterns during high-load hours, and to interactions among multiple binding constraints. Without attention, the model sees constraint data but cannot integrate it with temporal cues, and coverage drops 9.35 points with no change in point error. The near-redundancy of the autoregressive pathway (WOEnergyCancel) is the most theoretically significant result. When constraints and temporal features are present, lagged spreads add no detectable marginal calibration value. This contradicts the standard autoregressive paradigm—where past values of the target are assumed to carry predictive momentum—and supports the structural interpretation that spreads are congestion differentials. Once the congestion state is conditioned on, the realized spread is predictable without reference to its own history—a finding consistent with the market’s own structure: the day-ahead LMP spread is the output of a security-constrained optimization recomputed from scratch each day (stateless, with no carry-over memory), so modeling the congestion state that drives that optimization suffices. This finding echoes recent work showing that structural encodings can render autoregressive features redundant in other domains. **Comparison with Prior Work** The calibration-first evaluation philosophy aligns with emerging decision-aware paradigms in energy forecasting. [Hobbs et al., 2022] argue that probabilistic solar

forecasts should be scored by their operational value, and [Hirsch and Ziel, 2026] demonstrate that battery trading profits are more sensitive to interval calibration quality than to point forecast accuracy. SPARC’s advantage in coverage and Winkler score, with statistically indistinguishable CRPS versus linear baselines, provides empirical support for this philosophy: models optimized for point accuracy (LQR achieves CRPS 2.15) may be systematically overconfident, while calibration-optimized models sacrifice no meaningful point accuracy while gaining substantial distributional reliability. The cross-market-design transfer from European single-price imbalance to U.S. nodal day-ahead connects our work to the rule-embedding paradigm. [Yu et al., 2026] demonstrated that embedding settlement formulas as differentiable network layers improves European imbalance price forecasts. SPARC implements a complementary approach—conditioning on market-clearing outputs rather than embedding the clearing optimization—that transfers the principle to nodal markets where the clearing optimization is too complex for differentiable embedding. The ERCOT results (87.9% coverage) support the transfer hypothesis, though full bidirectional validation across multiple market designs remains future work. The constraint identity-versus-intensity decomposition echoes a broader theme in interpretable energy forecasting. [Toubeau et al., 2021] used attention mechanisms to identify which physical grid constraints drive forecast revisions, finding that qualitative constraint patterns explain forecast structure. SPARC’s double ablation (WOID vs. WOMu) provides a controlled experimental demonstration of this principle: identity ablation causes a 7.6-point coverage drop while intensity ablation causes a 1.0-point drop, establishing causal evidence rather than correlational attribution. **Practical Implications** SPARC’s parameter efficiency (8,978) and CPU-only training directly enable the operational rolling-window paradigm. Electricity markets experience structural change—generator retirements, transmission upgrades, seasonal load evolution—that erodes model calibration over months. The cross-year degradation results confirm this erosion empirically. A model that can be retrained monthly on the most recent two years of data at negligible computational cost can track regime changes, whereas static models or large architectures requiring GPU clusters cannot. The seasonal OOD results—strong near-term transfer (87.8% for January–May) with summer degradation—suggest that bi-annual retraining would capture the primary regime shifts, though summer extremes remain challenging. These two properties are use-case-contingent strengths rather than limitations for a hedging desk: cheap, CPU-only retraining is precisely the capability needed to track rolling market regimes, and a calibration-first

objective (reliable intervals at the quoted level) is the decision-relevant quantity that risk positions are sized from. Tuning the architecture toward point-error instead of interval reliability would sacrifice exactly the property that matters operationally; point-accuracy parity with the linear baseline is therefore not a short-fall but the deliberate cost of best-in-class interval reliability. The FERC Order 881 context extends SPARC’s applicability beyond ERCOT. All U.S. ISOs/RTOs now publish binding constraint data in standardized formats. While the constraint identities and market designs differ across regions, the architecture is market-agnostic: the constraint embedding table is repopulated with region-specific corridor identifiers, and the remainder of the model is unchanged. Cross-ISO deployment would require per-region constraint vocabulary mapping and potentially region-specific hyperparameter tuning, but the architectural principle—conditioning on published market-clearing outputs—transfers directly.

7 Limitations

Spike Coverage Is Insufficient. SPARC achieves 55.9% empirical coverage on the most extreme 5% of hours by absolute spread magnitude, substantially below the 87.9% global coverage. The model is poorly calibrated on tail events, which are precisely the hours where large congestion charges materialize. This limits SPARC’s applicability to routine hedging (the central 10–90% band) rather than tail-risk management. Extreme-value methods or models with explicit heavy-tailed likelihoods are more appropriate for spike forecasting. **Summer Degradation Is Structural.** The June–August seasonal window sees SPARC coverage fall to 62.5%, and all baselines degrade similarly. The most economically consequential season in ERCOT—when scarcity pricing, emergency reserves, and extreme congestion occur—is precisely where constraint-attention provides the least absolute calibration benefit. This degradation likely stems from temperature-driven transmission derating, scarcity pricing regimes, and binding constraint patterns that differ fundamentally from the rest of the year, making historical constraint patterns insufficient for reliable forecasting in this window.

8 Reproducibility and Data Availability

All models, splits, and results are reproduced from fixed seeds 42,43,44,45,46 on commodity CPU; no GPU was used. The full package is open source at the public repository, including:

- **Code:** model definitions, data loaders, runners

(main experiment, seasonal/cross-year OOD), calibration and coherence modules.

- **Per-seed outputs:** 334 artifacts (prediction, target, attention, and shift-factor `.numpy` arrays) for every condition and seed on the primary pair, plus 5-seed generalization-probe outputs and per-seed arrays on the two additional pairs.
- **Aggregate results:** machine-readable `results.json` with 5-fold per-seed statistics and statistical-test outputs.
- **Reproducibility entry point:** a single command reproduces the locked results from the raw parquet sources without retraining.

To ensure full transparency of the reported metrics, settings such as data provenance, chronological split boundaries, and metric definitions are version-controlled alongside the code. **Scoped-out areas.** Three directions surfaced during internal review were deliberately left out of this study and are listed here for completeness rather than as omissions: (1) validation across additional ISOs (e.g., PJM/CAISO) or market designs, which would strengthen generality but is left to future work; each ISO maintains its own constraint vocabulary and market-clearing format (not a standardized schema across regions), so cross-region deployment requires per-ISO constraint mapping; (2) a dedicated spike-calibration method, since the central-band focus is the scoped axis of this work and tail calibration is reported honestly as a limitation; and (3) a downstream hedging/trading simulation, which is intentionally excluded because this paper does not make trading-profit claims. None of these reflect hidden weaknesses; they are explicit boundaries of the presented contribution.

9 Conclusion

SPARC demonstrates that encoding market-clearing constraint signals through constraint-slot attention substantially improves probabilistic spread forecast calibration in U.S. nodal electricity markets. The model achieves the highest interval reliability among eight tested methods while maintaining competitive point accuracy and near-perfect quantile coherence, using fewer than 9,000 parameters trainable in minutes on commodity hardware. Ablation experiments isolate constraint-attention as the specific mechanism driving calibration gains, with constraint identity emerging as the dominant signal. The finding that autoregressive spread history is redundant when constraints are available challenges the standard time-series paradigm and supports a structural interpretation of spreads as congestion differentials. These results extend the

market-rule-embedding principle from European single-price imbalance markets to the structurally distinct U.S. nodal day-ahead setting, using real constraint data now publicly available under FERC Order 881. The calibration-first evaluation philosophy—leading with interval reliability rather than point accuracy—reflects the operational needs of congestion hedging, where position sizing depends on quantile estimates rather than mean forecasts. Future work should pursue three directions. First, validation across additional U.S. nodal markets (PJM, CAISO, MISO) and European flow-based market coupling regimes would test the generalizability of constraint-attention across diverse congestion architectures. Second, integrating antecedent scenario predictors—day-ahead weather forecasts, load forecasts, and generator commitment schedules—alongside constraint signals could address the documented summer degradation and improve extreme-hour calibration. Third, combining SPARC’s constraint-aware architecture with explicit tail-modeling approaches (extreme-value theory, normalizing flows for heavy-tailed distributions, or conformal prediction for tail-adaptive coverage) may bridge the gap between routine hedging reliability and spike-hour coverage, enabling a unified framework for both central-band and tail-risk forecasting.

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